



Emotional brain responses and pattern expressions in right temporal variant frontotemporal dementia

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Received: 29 July 2026 / Revised: 28 August 2026 / Accepted: 1 September 2026
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Dear Sirs,

Right temporal variant frontotemporal dementia (rtvFTD) is a distinctive clinical syndrome characterized by predominant right anterior temporal lobe degeneration and early socio-emotional symptoms, including impaired recognition of facial affect [1–3]. Recent task-based functional MRI work in behavioral variant FTD demonstrates reduced neural responses during affective empathy that correlate with informant-rated empathic concern, underscoring the feasibility and clinical relevance of probing socio-emotional mechanisms in vivo [4]. Although the behavioral phenotype is increasingly well delineated, the functional neural correlates that contribute to impaired processing of socio-emotional cues in rtvFTD remain poorly characterized. This gap is clinically relevant as emotion recognition deficits are

frequently reported early and contribute substantially to disrupted interpersonal functioning.

The anterior temporal lobe, particularly the temporal pole, has been implicated in integrating perceptual inputs with conceptual and affective meaning [5]. In the context of socio-emotional signals, this region is well positioned to link perceptual representations of faces to distributed representations of negative affect and social meaning. From a systems perspective, models that conceptualize the anterior temporal lobe as a transmodal integration hub predict that focal degeneration may be accompanied by both altered local responsivity to emotionally salient facial cues and distributed changes in affect-related representations across the broader face-responsive network [6]. However, direct task-based evidence for the coexistence of such regional and distributed functional alterations in rtvFTD remains scarce.

Here, we used task-based functional magnetic resonance imaging during perception of dynamic fearful and neutral facial expressions, with spatio-temporally scrambled control stimuli, in seven rtvFTD patients and nineteen healthy controls.

We focused on fear as a model of negative emotional meaning because the right anterior temporal region and its closely connected limbic structures, particularly the amygdala, have been consistently implicated in the detection, appraisal, and interpretation of threat-related facial signals. Fearful expressions therefore provided a theoretically motivated probe of a function likely to depend on the structural and functional integrity of the right anterior temporal network. Furthermore, because the clinical phenotype of rtvFTD limits the duration and complexity of task-based fMRI acquisition, we prioritized signal-to-noise ratio and the reliability of condition-specific estimates over emotion diversity. Including fewer conditions allowed more repetitions per condition and therefore more stable estimates of fearful versus neutral responsivity, while reducing participant burden and the risk of motion,

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fatigue, and incomplete data. Accordingly, fear was not selected on the assumption that it is uniquely impaired in rtvFTD, but as a biologically and clinically relevant model of negative facial-expression processing that could be examined with sufficient statistical reliability in this rare and challenging patient population.

We tested the *a priori* hypothesis that patients would show altered regional fearful versus neutral activation. To further probe network-level alterations, we complemented univariate analyses with multivariate signature-expression analyses [7] that quantify the expression of independently validated neural signatures associated with negative affect and related functions within a face-responsive network. Because the picture-induced negative affect signature was derived from responses to diverse aversive scenes rather than facial expressions, it was treated as a cross-context probe of distributed negative-affect engagement rather than as a face- or fear-specific marker. Accordingly, our inferences concern recognition and affective-semantic interpretation of fearful expressions rather than emotion processing more broadly.

This combined approach was intended to characterize complementary regional and distributed functional alterations associated with impaired interpretation of fearful facial expressions in rtvFTD.

The protocol was approved by the UZ Leuven ethical committee and all participants provided informed consent prior to participation. 20 patients with behavioral variant FTD (bvFTD) and 20 healthy controls participated in the original study cohort, which has been reported in detail elsewhere [8]. One control and one patient were excluded because of technical issues. To identify patients with rtvFTD, we followed a procedure similar to that described by Younes et al. [3] based on *W* scores of regional atrophy patterns. T1-weighted MRI images were processed with FreeSurfer 6.0 using the default recon-all pipeline, and regional gray matter volumes were extracted according to the Desikan–Killiany–Tourville atlas [9, 10]. *W* scores were calculated as (observed gray matter volume – predicted gray matter volume)/root mean square error. Predicted regional volumes and root mean square error were derived from a normative sample of 2757 controls using linear regression models adjusted for age, sex, total intracranial volume, and scanner manufacturer, following Potvin et al. [11]. A right temporal index was calculated as mean frontal *W* score/mean right temporal *W* score. Patients were classified as rtvFTD when (i) their three lowest regional *W* scores were located in the right temporal lobe and (ii) the right temporal index was < 0.50 . This procedure identified seven patients with rtvFTD (Fig. 1). Behavioral symptoms were retrospectively coded from electronic health records, integrating neuropsychological reports and specialist consultation notes, to determine the presence of the three core rtvFTD symptom clusters [2] (Table 1).

Participants viewed 2 s videos of fearful faces, neutral faces, and spatio-temporally scrambled versions in an event-related oddball paradigm. Post-scanning, participants rated each face stimulus for valence, arousal, and emotion category. For task details, see [8].

Preprocessing via fMRIPrep included MNI normalization, motion correction, and co-registration, followed by 8-mm spatial smoothing [12]. GLM-based first-level analyses focused on two contrasts: faces $>$ scrambled stimuli and fearful $>$ neutral faces. To characterize which condition contributed to any fearful $>$ neutral group difference, mean parameter estimates for fearful $>$ baseline and neutral $>$ baseline were extracted separately from the identified cluster. Because these estimates were extracted from a region defined by the same contrast being decomposed, they are presented descriptively and were not subjected to additional inferential testing.

For the whole-brain analyses, the primary exploratory threshold was an uncorrected voxel-wise $p < 0.001$ combined with a minimum cluster extent of $k \geq 50$ voxels. This threshold was selected to balance sensitivity and specificity in the context of the small sample available for this rare clinical phenotype and aligns with cluster-defining threshold of similar studies with substantially larger samples [13]. We therefore considered the present threshold appropriate for exploratory localization, but not for confirmatory inference.

As a more stringent secondary analysis, all whole-brain contrasts were additionally evaluated using false discovery rate (FDR) correction at $q < 0.05$. Findings present only at the uncorrected threshold were interpreted as preliminary and hypothesis-generating.

For uncorrected group differences, a standardized effect size (Hedges' *g*) was computed from each participant's mean contrast value within the identified cluster using the pooled-variance formula with a small-sample bias correction, and 95% confidence intervals were obtained using bias-corrected and accelerated bootstrapping with 5000 resamples, independently resampling each group. Because these effect sizes were extracted from clusters defined by the same group contrast, they are reported descriptively and should not be interpreted as unbiased estimates of the population effect.

For multivariate analyses, beta maps for fearful faces versus fixation and neutral faces versus fixation were extracted and signature expression was quantified using cosine similarity within a face-responsive region of interest (ROI). The signatures indexed picture-induced negative affect, subjective fear, empathic care, and empathic distress [14–16]. The face-responsive ROI was defined by binarizing the faces $>$ scrambled contrast across all 26 participants (one-sample *t*-test, $p < 0.001$). These analyses were considered exploratory because multiple independently derived signatures were evaluated and because the signatures were developed using paradigms that differed from the present facial-expression

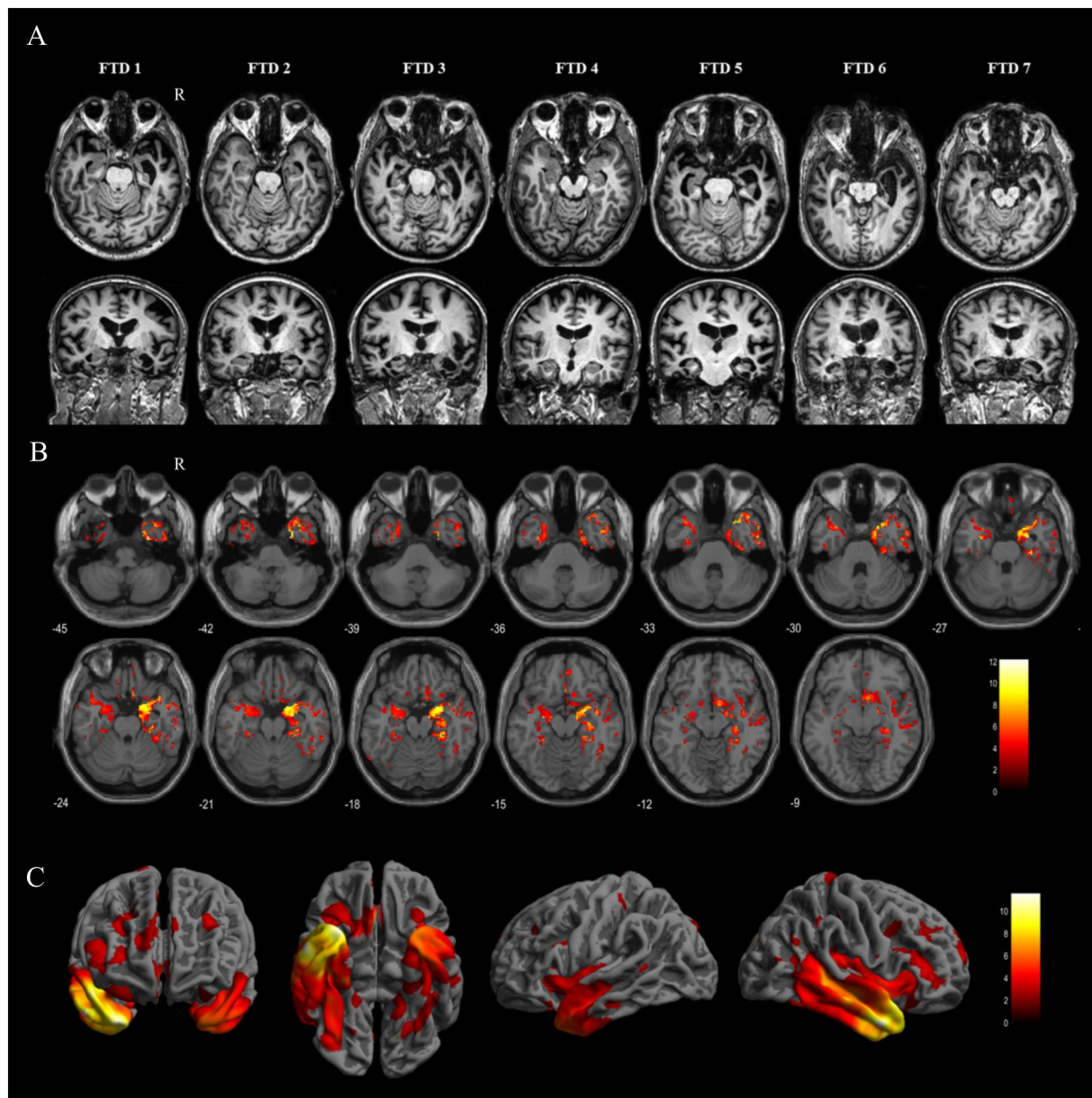


Fig. 1 Example slices and structural group differences. **A** Sagittal and coronal slices for each rlvFTD participant illustrating right anterior temporal lobe atrophy. **B** Group difference in gray matter volume

based on a two-sample *t*-test in CAT12 following preprocessing with default settings (FDR-corrected, $q < 0.05$). **C** Group difference in cortical thickness (FDR-corrected, $q < 0.05$)

Table 1 Demographic and clinical characteristics of the sample

	Controls	rtvFTD	<i>p</i> Group difference
Age (years, Mean/Std)	66.32/6.14	65.91/8.63	0.82 (Mann–Whitney <i>U</i>)
Sex (female/male)	8/11	4/3	0.913 (Chi-square)
MMSE (Mean/Std)	29.26/0.26	26.71/1.98	0.002* (Mann–Whitney <i>U</i>)
Disease duration (Mean/Std in years)	NA	1.9/0.93	NA
Diagnostic criteria			
Socio-emotional disturbances	NA	7/7	NA
Impaired non-verbal multimodal knowledge	NA	6/7	NA
Focus on specific interest	NA	4/7	NA

Std standard deviation, MMSE Mini-Mental State Examination, NA not applicable

task. Group differences were examined using Mann–Whitney U and permutation tests. Bonferroni correction was applied across the eight signature-by-condition comparisons, while uncorrected results are additionally reported as exploratory findings. Rank-biserial correlations were used as standardized effect sizes, with 95% confidence intervals obtained using bias-corrected and accelerated bootstrapping with 5000 resamples. Spearman correlations tested brain–behavior associations within the rtvFTD sample, with 95% confidence intervals obtained using the same bootstrap procedure.

The primary behavioral hypothesis concerned impaired categorization of fearful facial expressions in rtvFTD. Categorization accuracy, valence, and arousal ratings were compared between groups for fearful and neutral faces using Mann–Whitney U tests. Bonferroni correction was applied across the six behavioral comparisons. Directional one-tailed tests were used for the pre-specified hypotheses of poorer fearful-face categorization and reduced neural engagement in rtvFTD. Given the small sample size and absence of preregistration, the corresponding two-tailed p values were additionally examined as sensitivity analyses and are reported alongside the directional findings.

At the uncorrected level, controls showed higher fearful-face categorization accuracy than rtvFTD patients (Mann–Whitney $U = 32$, one-tailed $p = 0.021$, corresponding two-tailed $p = 0.042$; $r_{rb} = 0.52$, BCa 95% CI [0.03, 0.82]) and rated fearful faces as more negative ($U = 34$, one-tailed $p = 0.031$, corresponding two-tailed $p = 0.062$; $r_{rb} = 0.49$, BCa 95% CI [−0.17, 0.84]).

Fearful-face arousal ratings did not differ significantly between groups ($U = 46$, one-tailed $p = 0.123$, corresponding two-tailed $p = 0.246$; $r_{rb} = 0.31$, BCa 95% CI [−0.33, 0.74]), and no group differences were observed for neutral-face ratings. None of the six behavioral comparisons remained significant after Bonferroni correction. Accordingly, the uncorrected difference in fearful-face categorization is considered preliminary evidence consistent with the a priori directional hypothesis, whereas the valence effect did not remain significant in the two-tailed sensitivity analysis.

The wide confidence intervals, particularly for valence, underscore the imprecision of effect-size estimation in this small sample (Fig. 2A, C).

No group differences survived whole-brain FDR correction at $q < 0.05$ for either contrast. At the pre-specified exploratory uncorrected threshold of $p < 0.001$ and $k \geq 50$ voxels, the fearful > neutral contrast identified a right lateral temporal pole cluster in which controls showed higher signed fearful versus neutral contrast values than rtvFTD patients (peak MNI: 55, 11, −12; $Z = 4.70$; $k = 169$). The standardized effect size within this cluster was large (Hedges' $g = 2.52$, 95% CI [1.29, 3.63]) although this estimate is descriptive because the same contrast was used

to define the cluster and estimate the effect. No significant group difference was detected for the faces > scrambled contrast at either threshold. This absence of a detected difference should not be interpreted as evidence of equivalence between groups.

Descriptive examination of the condition-specific parameter estimates showed that the fearful > neutral group difference reflected opposite response profiles rather than a simple reduction in differentiation (Fig. 3B). Relative to fixation, controls showed positive mean activation to fearful faces and negative mean activation to neutral faces, whereas rtvFTD patients showed negative mean activation to fearful faces and positive mean activation to neutral faces. The absolute separation between fearful and neutral responses was therefore not reduced in rtvFTD and was, descriptively, larger than in controls, but its direction was reversed. The group effect is consequently better characterized as reversed condition-specific modulation of right temporal pole activity by fearful versus neutral facial expressions than as an isolated reduction in fear-related activation. Because these estimates were extracted from the cluster identified by the same fearful > neutral group contrast, they are presented descriptively and were not subjected to additional inferential testing.

Picture-induced negative affect, subjective fear, empathic care, and empathic distress signatures covered 100%, 97.4%, 60.2%, and 74.6% of the face-responsive ROI, respectively. At the uncorrected level, controls showed a higher expression of the picture-induced negative affect signature during fearful-face viewing than rtvFTD patients (Mann–Whitney $U = 32$, one-tailed $p = 0.025$, corresponding two-tailed $p = 0.050$; permutation $p = 0.041$, corresponding two-tailed $p = 0.082$; $r_{rb} = 0.52$, BCa 95% CI [−0.24, 0.87]). This difference did not remain significant after Bonferroni correction across the eight signature-by-condition comparisons (adjusted $p = 0.200$ for the Mann–Whitney test; adjusted permutation $p = 0.328$). No other signature showed a group difference for fearful or neutral faces (see Fig. 4). Accordingly, reduced picture-induced negative affect signature expression is interpreted as an exploratory finding, with the direction of the effect consistent across the rank-based and permutation analyses.

Within the rtvFTD group, activation for fearful faces versus fixation correlated positively with arousal ratings for fearful faces (Spearman $r_s = 0.775$, $p = 0.020$, BCa 95% CI [0.19, 1.00], based on 5000 bootstrap samples). Given $n = 7$, this association is considered exploratory.

The present exploratory findings suggest altered, and descriptively reversed, modulation of right temporal pole activity by fearful versus neutral facial expressions in rtvFTD. No whole-brain group difference survived FDR correction, but at the prespecified uncorrected threshold a right lateral temporal pole cluster showed higher signed fearful versus neutral contrast values in controls than in patients.

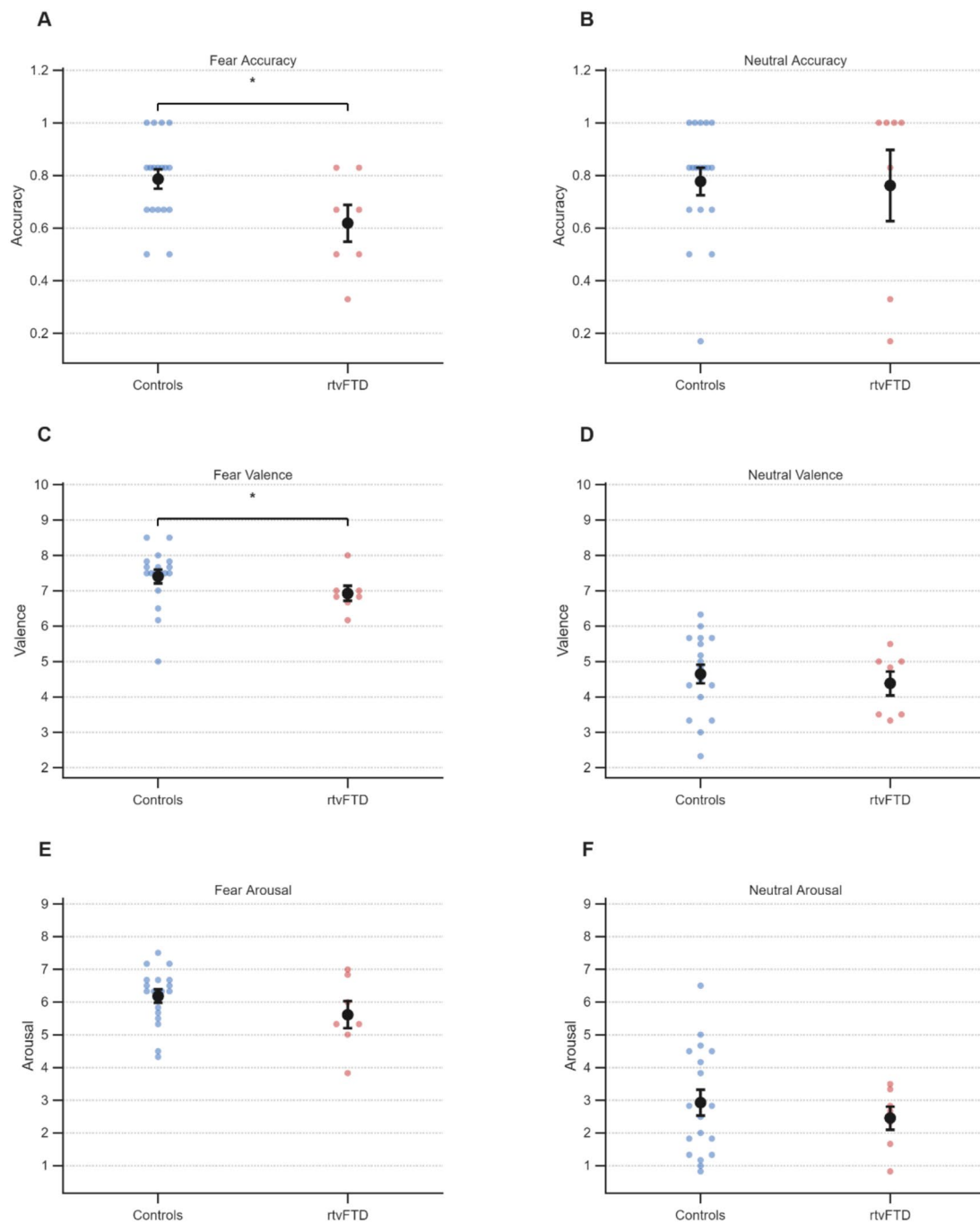


Fig. 2 Group comparison of post-scanning ratings. **A** categorization fearful faces, **B** categorization neutral faces, **C** valence rating fearful faces (rated from 1 to 9 corresponding from highly positive to highly negative), **D** valence rating neutral faces, **E** indicated arousal levels

when viewing fearful faces, **F** indicated arousal levels when viewing neutral faces. * $p < 0.05$. Error bars represent group means $\pm$ standard error of the mean. * $p < 0.05$, uncorrected

Descriptive decomposition revealed opposite condition-specific response profiles: controls showed higher activation to fearful than neutral faces, whereas rtvFTD patients showed higher activation to neutral than fearful faces. Importantly, the absolute separation between the two conditions was not reduced in rtvFTD and was descriptively larger than

in controls. The finding is therefore better characterized as reversed affective modulation of right anterior temporal activity than as an attenuation of fear-related responsivity. No significant group difference was detected for faces versus scrambled stimuli although this does not establish equivalence of general face processing between groups.

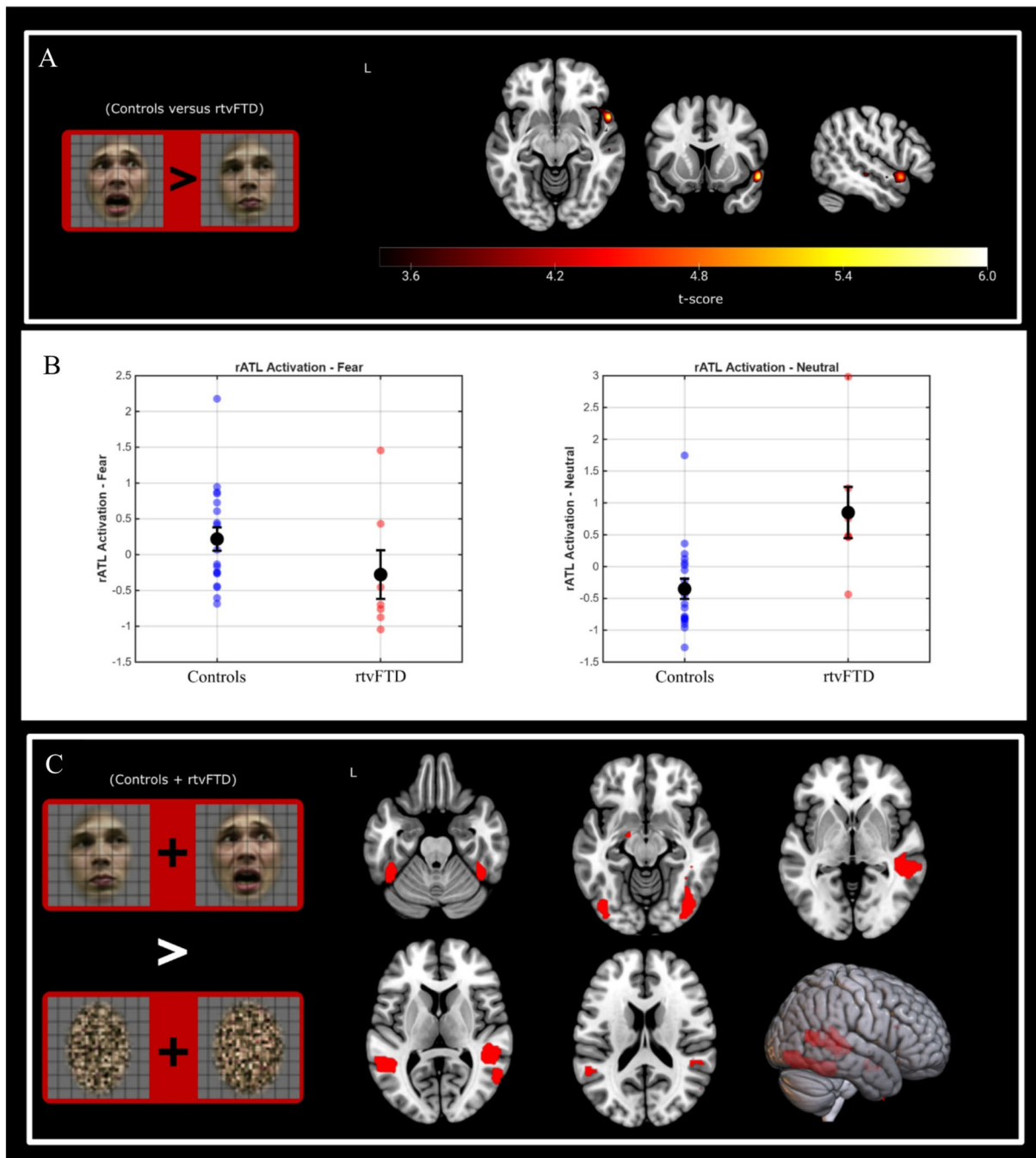


Fig. 3 Imaging results. **A** Z-map of fearful > neutral group difference. **B** Mean activation for fearful and neutral faces relative to baseline within the right lateral temporal pole cluster identified by the exploratory fearful > neutral group contrast. Controls showed relatively greater activation to fearful than neutral faces, whereas rtvFTD patients showed the opposite profile. The absolute separation between

conditions was descriptively larger in rtvFTD, but reversed in direction. Values are shown descriptively because the cluster was defined by the same contrast being decomposed. Error bars represent 95% confidence intervals. **C** Overall faces > scrambled thresholded at $Z > 3.09$ to obtain a ROI mask of 955 voxels. *L* left hemisphere

This interpretation is compatible with semantic neuro-anatomical accounts in which posterior visual systems support perceptual analysis while the anterior temporal lobes integrate perceptual inputs with conceptual knowledge.

Dual-route models further assign the right anterior temporal lobe a particular role in non-verbal access to meaning [17]. The reversed condition-specific response profile may therefore reflect abnormal mapping of visual facial information

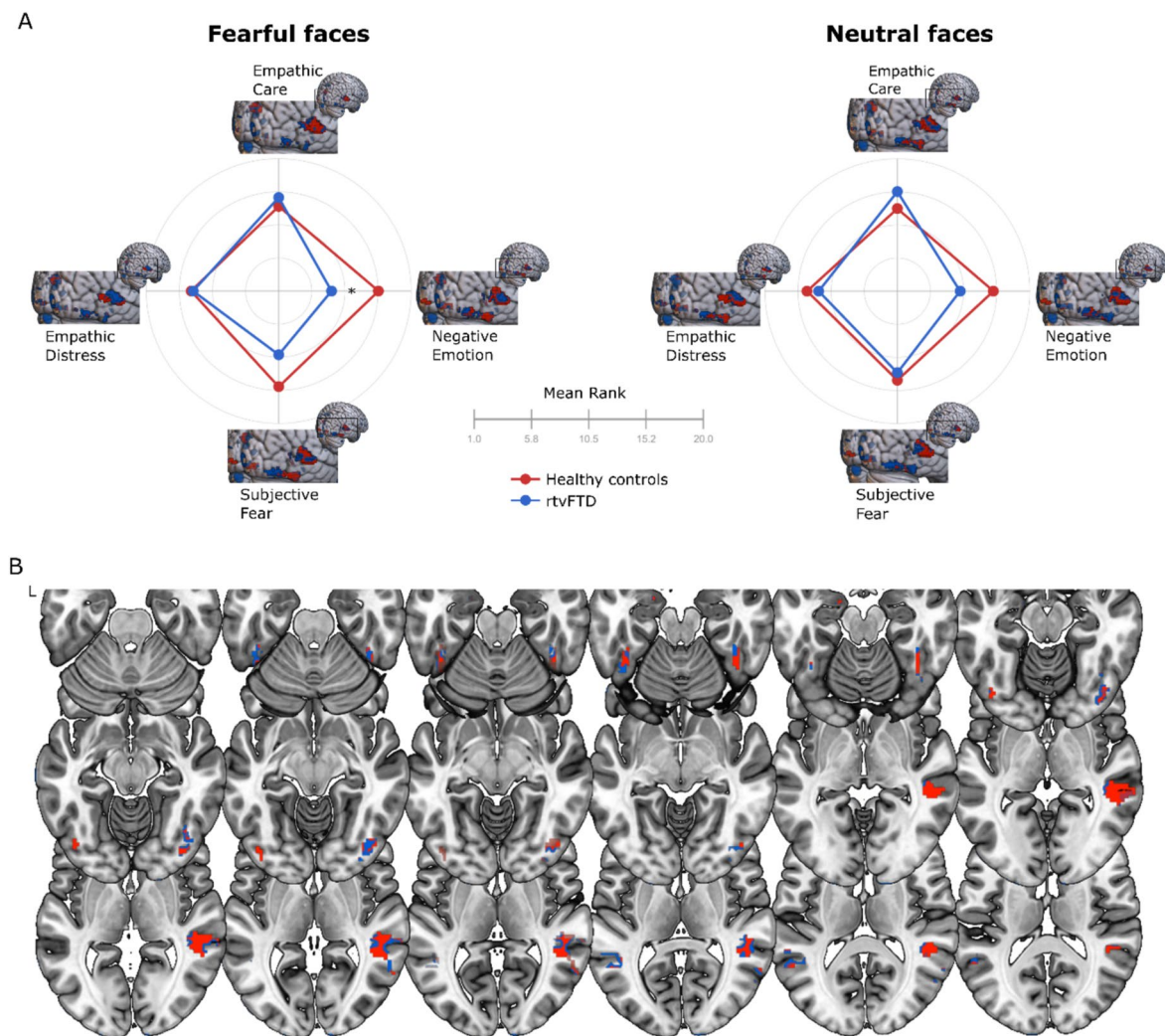


Fig. 4 Signature expression group comparison. **A** Radar plot displaying overall similarity between brain responses and signatures of empathic care, picture-induced negative affect, subjective fear, and empathic distress during perception of fearful (left) and neutral (right) faces. Brain maps depict where controls (red) or rtvFTD patients (blue) exhibit higher signature expression within the ROI.

Comparison is based on mean ranks, consistent with Mann–Whitney U methodology. $*p < 0.05$, uncorrected. **B** Axial slices showing where in the face network controls (red) or rtvFTD patients (blue) exhibit higher picture-induced negative affect signature expression during fearful-face perception. *L* left hemisphere

onto affective-semantic meaning rather than merely weaker neural engagement by fearful expressions. The lower post-scan accuracy for fearful-expression categorization is not necessarily a confound that can be separated from the neural effect. Rather, it may reflect the same difficulty in extracting or accessing the emotional–semantic meaning of the stimuli. Importantly, participants passively viewed the stimuli during fMRI and did not perform an online categorization task. Thus, the temporal pole difference cannot be attributed to failure to execute an in-scanner task although the present data cannot determine whether the reversed modulation reflects reduced access to fearful meaning, altered appraisal of neutral expressions, or a combination of both. The correlation between temporal pole responsivity and arousal

ratings provides tentative behavioral relevance but should be interpreted cautiously given the small patient sample.

The multivariate analyses provided a complementary, but also exploratory, pattern. At the uncorrected level, rtvFTD patients showed reduced expression of the picture-induced negative affect signature during fearful-face viewing. This result did not survive Bonferroni correction and the two-tailed sensitivity analysis was a borderline for the rank-based test and nonsignificant for the permutation test. Moreover, the signature was developed to predict negative affect elicited by aversive scenes, such as injury, aggression, accidents, and contamination, rather than by facial expressions or fear-specific paradigms. Viewing another person's fearful expression and experiencing negative affect in response to

an aversive scene involve overlapping but non-equivalent processes. Reduced signature expression therefore does not demonstrate an impaired neural representation of fear. More cautiously, it suggests reduced engagement of a distributed pattern associated across contexts with negative affect. The absence of a group difference for the subjective-fear signature further argues against a fear-specific interpretation.

Across behavioral, regional, and multivariate analyses, the direction of the findings was broadly consistent with altered processing of fearful relative to neutral facial expressions. The regional effect is particularly notable because the two conditions remained clearly differentiated in rtvFTD, but in the opposite direction from controls, arguing against a simple loss of condition sensitivity. However, these analyses were conducted in the same small cohort and should not be considered statistically independent evidence of a single mechanism. We therefore interpret the regional temporal pole finding and distributed signature result as complementary observations rather than as evidence for directly coupled abnormalities.

Within hub-and-spoke and graded-specialization accounts, fearful expressions can be considered non-verbal inputs whose significance depends on anterior temporal integration with distributed sensory, conceptual, and affective systems [6, 17]. The reversed temporal pole response profile may therefore index an alteration in how visual facial features are mapped onto emotion concepts or affective significance, whereas the reduced expression of the picture-induced negative affect pattern may reflect weaker recruitment of distributed systems associated with negative affect. This framework accommodates the regional and distributed observations without assuming that the temporal pole contains a unitary representation of fear or that the two imaging measures are causally linked. It also provides a neuro-functional account of why patients may detect a face yet have difficulty identifying, interpreting, or appraising its fearful expression, consistent with neuropsychological observations in right anterior temporal degeneration [3] and Mesulam's framework linking percepts to experiential meaning [18].

We did not observe group differences in signatures of subjective fear, empathic care, or empathic distress. These null results may reflect construct–task correspondence and limited statistical power. Passive viewing of brief facial expressions may not reliably elicit the internal-state components captured by these signatures, and the signatures were derived from paradigms that differed substantially in stimulus modality and psychological demands. Accordingly, the positive picture-induced negative affect result should be interpreted as cross-context evidence for altered negative-affect engagement rather than as a biomarker of fear recognition or subjective fear in rtvFTD.

Several limitations warrant consideration. First, the sample was small, reflecting both the rarity of rtvFTD and the

practical difficulty of recruiting and retaining patients with this behavioral phenotype for demanding task-based fMRI protocols. We therefore used an exploratory uncorrected voxel-wise threshold intended to preserve sensitivity and reduce Type II error, consistent with sensitivity-oriented cluster-defining approaches used in prior FTD fMRI work [13]. Nevertheless, no whole-brain result survived FDR correction, and neither the behavioral nor multivariate findings survived correction for their respective numbers of comparisons. The principal findings should therefore be regarded as preliminary and hypothesis-generating. Although several point estimates were large, confidence intervals were wide, reflecting substantial uncertainty about effect magnitude. Replication in larger, preferably multicenter cohorts is required.

Second, the principal functional effect was located within the markedly atrophic right anterior temporal lobe. Because the critical effect was based on the within-subject fearful > neutral contrast, static reductions in BOLD signal related to tissue loss, partial-volume effects, or generally reduced signal-to-noise would be expected to affect both conditions and therefore be attenuated by subtraction. The condition-specific estimates further argue against a mere global signal-loss account: rtvFTD patients did not show uniformly reduced responses across both conditions, but instead showed a descriptively larger separation between fearful and neutral responses than controls, with the direction of that separation reversed. Nevertheless, subtraction cannot eliminate all consequences of neurodegeneration: atrophy may alter neurovascular responsiveness or BOLD dynamic range, and smoothing may interact with condition-specific activation patterns. Covarying regional gray matter volume out could remove disease-relevant variance and lead to over-adjustment because local atrophy is intrinsic to the rtvFTD phenotype and likely biologically related to the functional alteration of interest. The finding should consequently be interpreted as reversed fearful versus neutral responsivity within an atrophic right temporal region rather than as proof of a functional abnormality independent of structural degeneration.

Third, we examined only one negative emotion category. Fear was selected as a theoretically motivated model of negative facial-expression processing and to maximize signal-to-noise ratio within the practical constraints of fMRI in this clinical population. We did not assume that fear is uniquely vulnerable in rtvFTD. The findings therefore cannot establish whether the observed abnormalities are specific to fear or generalize to other negative or complex emotions. Fourth, no disease comparison groups were included, so syndrome specificity cannot be established. Direct comparison with frontal-predominant bvFTD and left-predominant semantic variant primary progressive aphasia would help distinguish effects related to right temporal topography from those

shared across FTD syndromes. Finally, signature expression was quantified within a face-responsive mask derived across all participants. Future studies should evaluate robustness to alternative mask definitions and analytic choices.

In conclusion, this small exploratory study identified reversed condition-specific modulation of right temporal pole activity by fearful versus neutral facial expressions in rtvFTD, together with reduced expression of a distributed pattern previously linked to picture-induced negative affect. The absolute separation between fearful and neutral responses was descriptively larger in rtvFTD than in controls, but occurred in the opposite direction, indicating altered affective modulation rather than a mere loss of differentiation. Neither of the imaging findings survived the most stringent correction procedures and both require replication. Nevertheless, the behavioral and imaging observations are compatible with the hypothesis that right anterior temporal degeneration alters the mapping of facial cues onto affective-semantic meaning.

Funding Jvds is supported by FWO-Vlaanderen (G030025N & G019625N), KU Leuven (IDN21/010), the King Baudouin Foundation (2023-J1990130-233837) and the Sequoia Fund for Research on Ageing and Mental Health.

Data availability The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available as they include information that could compromise the privacy of the research participants.

Declarations

Conflicts of interest The authors have no conflict of interest to report.

Ethical approval This study was conducted in compliance with the Declaration of Helsinki and approved by the ethical committee of University Hospitals Leuven, Belgium.

Informed consent All participants gave written informed consent.

References

1. Ulugut H, Bertoux M, Younes K, Montembeault M, Fumagalli GG, Samanci B, Illan-Gala I, Kuchcinski G, Leroy M, Thompson JC, Kobylecki C, Santillo AF, Englund E, Waldo ML, Riedl L, Van den Stock J, Vandenbulcke M, Vandenberghe R, Laforce R Jr, Ducharme S, Pressman PS, Caramelli P, de Souza LC, Takada LT, Gurvit H, Hansson O, Diehl-Schmid J, Galimberti D, Pasquier F, Miller BL, Scheltens P, Ossenkoppele R, van der Flier WM, Barkhof F, Fox NC, Sturm VE, Miyagawa T, Whitwell JL, Boeve B, Rohrer JD, Gorno-Tempini ML, Josephs KA, Snowden J, Warren JD, Rankin KP, Pijnenburg YAL, International rtv FTDWG (2024) Clinical recognition of frontotemporal dementia with right anterior temporal predominance: a multicenter retrospective cohort study. *Alzheimers Dement* 20:5647–5661
2. Ulugut H, Younes K, Montembeault M, Bertoux M, Irish M, Kumfor F, Fumagalli GG, Samanci B, Illan-Gala I, Thompson JC, Santillo AF, Englund E, Landqvist Waldo M, Riedl L, Van den Stock J, Vandenbulcke M, Vandenberghe R, Laforce R Jr, Ducharme S, Pressman PS, Caramelli P, Cruz de Souza L, Takada LT, Gurvit H, Diehl-Schmid J, Galimberti D, Pasquier F, Weintraub S, Miller BL, Sturm VE, Whitwell JL, Boeve B, Rohrer JD, Piguet O, Gorno-Tempini ML, Josephs KA, Snowden J, Rowe JB, Warren JD, Rankin KP, Pijnenburg YAL, International rtv FTDWG (2025) Clinical recognition of frontotemporal dementia with right temporal predominance: a consensus statement from the International Working Group. *Commun Med (Lond)* 5:523
3. Younes K, Borghesani V, Montembeault M, Spina S, Mandelli ML, Welch AE, Weis E, Callahan P, Elahi FM, Hua AY, Perry DC, Karydas A, Geschwind D, Huang E, Grinberg LT, Kramer JH, Boxer AL, Rabinovici GD, Rosen HJ, Seeley WW, Miller ZA, Miller BL, Sturm VE, Rankin KP, Gorno-Tempini ML (2022) Right temporal degeneration and socioemotional semantics: semantic behavioural variant frontotemporal dementia. *Brain* 145:4080–4096
4. Lindberg O, Li TQ, Lind C, Vestberg S, Almkvist O, Stiernstedt M, Ericson A, Bogdanovic N, Hansson O, Harper L, Westman E, Graff C, Tsevis T, Mannfolk P, Fischer H, Nilsson G, Petrovic P, Nyberg L, Wahlund LO, Santillo AF (2024) Altered empathy processing in frontotemporal dementia. *JAMA Netw Open* 7:e2448601
5. Rouse MA, Binney RJ, Patterson K, Rowe JB, Lambon Ralph MA (2024) A neuroanatomical and cognitive model of impaired social behaviour in frontotemporal dementia. *Brain* 147:1953–1966
6. Ralph MA, Jefferies E, Patterson K, Rogers TT (2017) The neural and computational bases of semantic cognition. *Nat Rev Neurosci* 18:42–55
7. Van Oudenhove L, Kragel PA, Dupont P, Ly HG, Pazmany E, Enzlin P, Rubio A, Delon-Martin C, Bonaz B, Aziz Q, Tack J, Fukudo S, Kano M, Wager TD (2020) Common and distinct neural representations of aversive somatic and visceral stimulation in healthy individuals. *Nat Commun* 11:5939
8. De Winter FL, Van den Stock J, de Gelder B, Peeters R, Jastorff J, Sunaert S, Vanduffel W, Vandenberghe R, Vandenbulcke M (2016) Amygdala atrophy affects emotion-related activity in face-responsive regions in frontotemporal degeneration. *Cortex* 82:179–191
9. Fischl B (2012) FreeSurfer. *Neuroimage* 62:774–781
10. Desikan RS, Segonne F, Fischl B, Quinn BT, Dickerson BC, Blacker D, Buckner RL, Dale AM, Maguire RP, Hyman BT, Albert MS, Killiany RJ (2006) An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *Neuroimage* 31:968–980
11. Potvin O, Dieumegarde L, Duchesne S, Alzheimer's Disease Neuroimaging I (2017) Freesurfer cortical normative data for adults using Desikan–Killiany–Tourville and ex vivo protocols. *Neuroimage* 156:43–64
12. Esteban O, Markiewicz CJ, Blair RW, Moodie CA, Isik AI, Erramuzpe A, Kent JD, Goncalves M, DuPre E, Snyder M, Oya H, Ghosh SS, Wright J, Durnez J, Poldrack RA, Gorgolewski KJ (2019) FMRIPrep: a robust preprocessing pipeline for functional MRI. *Nat Methods* 16:111–116
13. Marshall CR, Hardy CJD, Russell LL, Bond RL, Sivasathiaselan H, Greaves C, Moore KM, Augustus JL, van Leeuwen JEP, Wasling SJ, Rohrer JD, Kilner JM, Warren JD (2019) The functional neuroanatomy of emotion processing in frontotemporal dementias. *Brain* 142:2873–2887
14. Ashar YK, Andrews-Hanna JR, Dimidjian S, Wager TD (2017) Empathic care and distress: predictive brain markers and dissociable brain systems. *Neuron* 94:1263–1273 e1264
15. Chang LJ, Gianaros PJ, Manuck SB, Krishnan A, Wager TD (2015) A sensitive and specific neural signature for picture-induced negative affect. *PLoS Biol* 13:e1002180

16. Zhou F, Zhao W, Qi Z, Geng Y, Yao S, Kendrick KM, Wager TD, Becker B (2021) A distributed fMRI-based signature for the subjective experience of fear. *Nat Commun* 12:6643
17. Hurley RS, Mesulam MM, Sridhar J, Rogalski EJ, Thompson CK (2018) A nonverbal route to conceptual knowledge involving the right anterior temporal lobe. *Neuropsychologia* 117:92–101
18. Mesulam M (2008) Representation, inference, and transcendent encoding in neurocognitive networks of the human brain. *Ann Neurol* 64:367–378